

Building Style Transfer Tool from AI Models

OBJECTIVE

Programming is focused on building tools that could make artists' work more cohesive on both the front-end and backend of traditional pipelines

APPROACH

Evaluate both a reference image for style such as a painter or concept artist, and a second image to represent luminance value to transfer the style

AI MODELS / TECHNIQUE

- VGG 19
 - Extract semantic features from images at multiple levels of abstraction including edges and color, textures and patterns
 - Pre-trained AI model (no training required)
- AdaIN (Adaptive Instance Normalization)
 - Fast color and tone transfer
 - Require training (using MS-COCO Datasets)
- Tonal Mask Generation
 - Creates three smooth, overlapping masks based on image luminance
- Manual Mask Integration
 - User-defined regions for selective style application
- Claude AI for coding assistance



Auto-Generated Luminance Mask



Highlight



Midtone



Shadow

FINAL RESULT

Combine Three References



Highlight



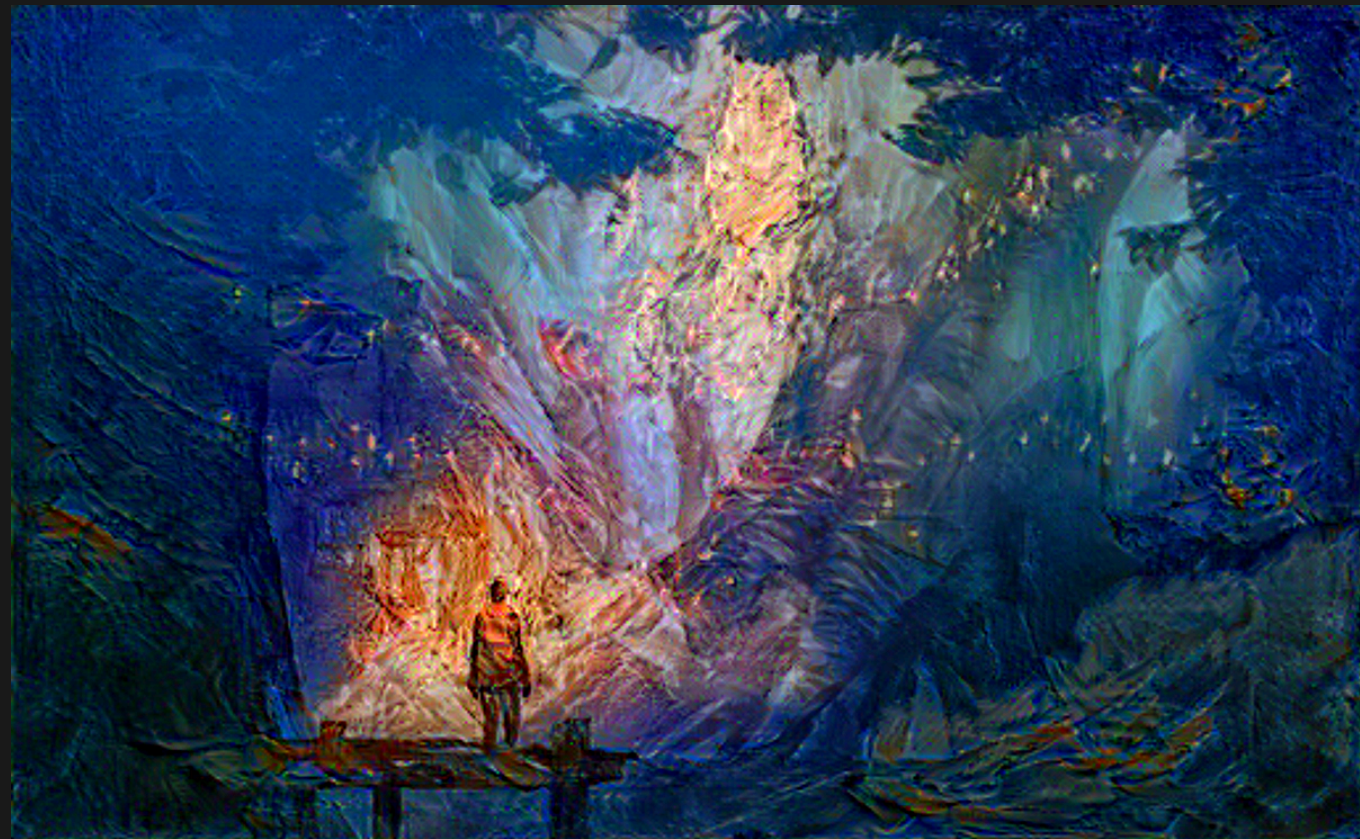
Midtone



Shadow

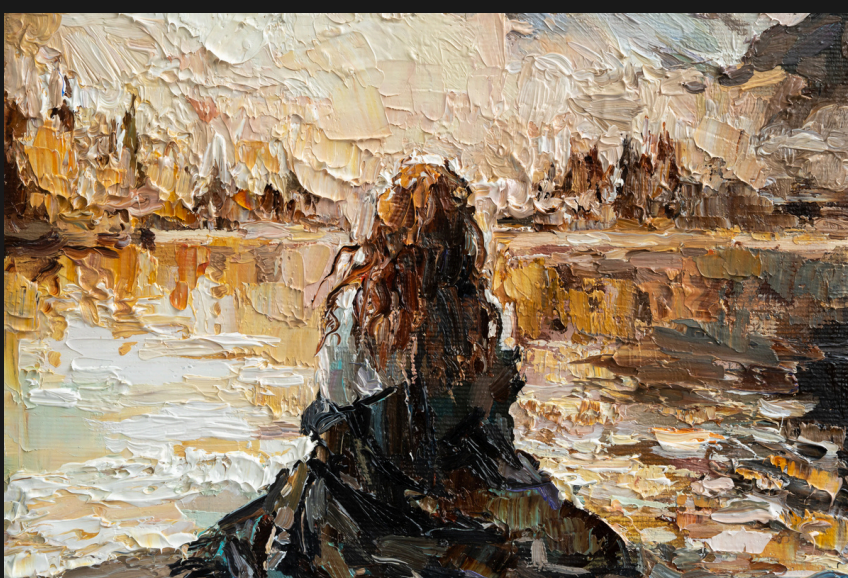


Original Image



Output

Custom Mask



Reference



Original Design



Mask

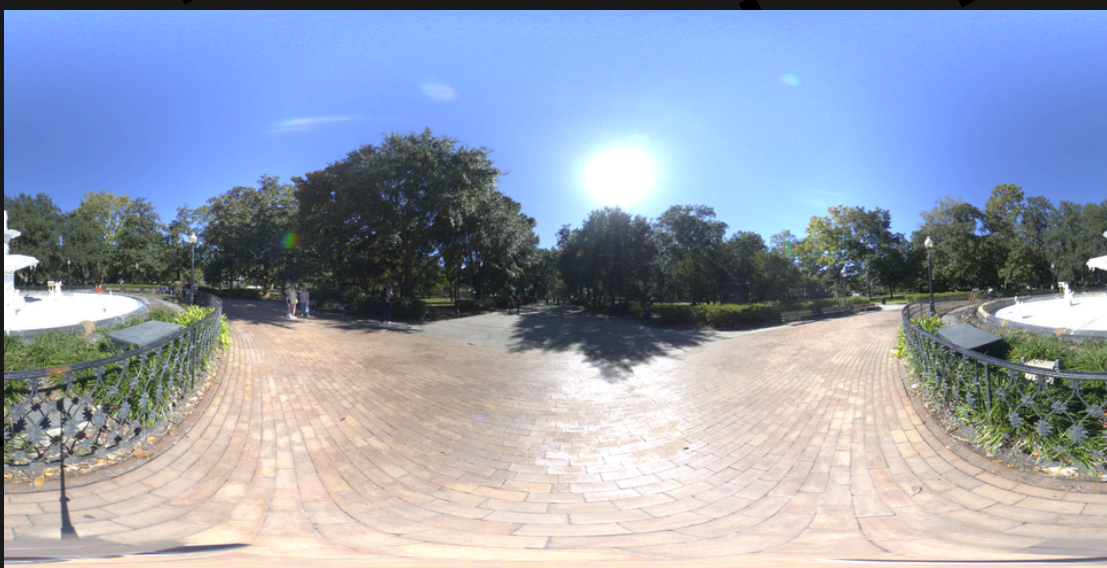


Effects With Mask

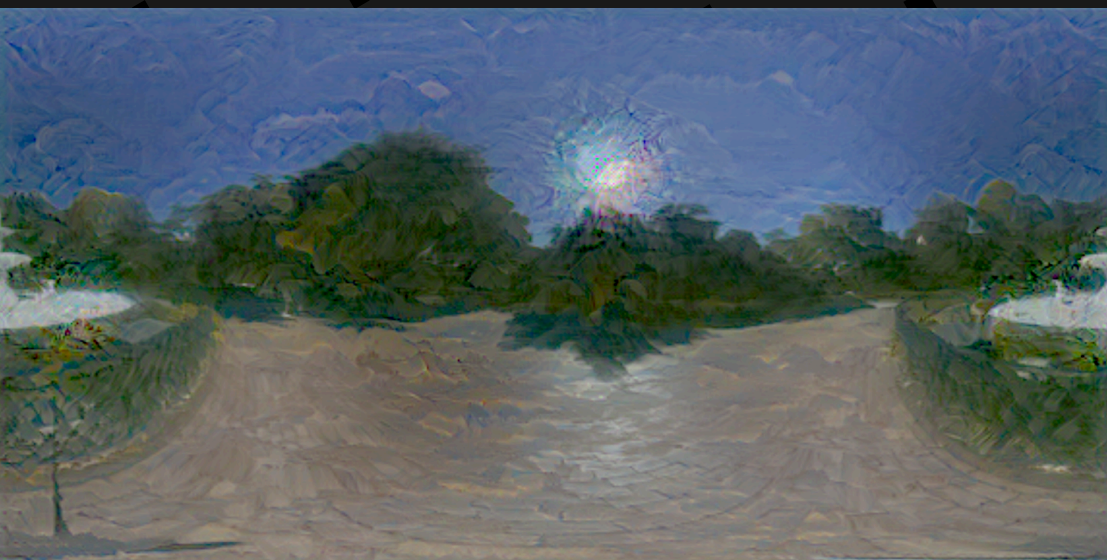


Effects Without Mask

Stylized HDRI



Original Image



Output



Test Render